Sai Ritwik Kosi Reddy

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SUMMARY

Cloud Automation Engineer specializing in Python-driven automation, Infrastructure as Code (Terraform, CloudFormation), and Ansible across AWS and Azure. Proven experience delivering large-scale infrastructure automation, including provisioning **700+ production servers** and designing enterprise-grade deployment workflows.
Focused on building reliable, secure, and scalable systems with strong emphasis on observability, governance, and operational excellence. Proven ability to automate complex infrastructure processes, resolve production issues, and deliver high-impact solutions in collaboration with DevOps and platform teams.

SKILLS

Automation & IaC: Terraform, Ansible, CloudFormation, AWS CDK, Infrastructure as Code (IaC), Configuration Management, Workflow Automation
Programming: Python, Go, PowerShell, Bash, SQL
Cloud & Infrastructure: AWS (EC2, ECS, S3, Lambda, IAM, RDS, VPC), Azure, Kubernetes (AKS, EKS), Docker
CI/CD & DevOps: GitHub Actions, Jenkins, Azure DevOps, OIDC Authentication, Pipeline Automation
Monitoring & Observability: CloudWatch, Azure Monitor, Log Analytics, Logging & Alerting
Governance & Security: IAM, Access Control, Resource Tagging, Policy Enforcement

WORK EXPERIENCE

Software Infrastructure Engineer Infinite Computer Solutions (Client: Molina Healthcare) Missouri, USA
Sep 2024 – Present

- Provisioned **700+ production-grade servers** using Terraform and Ansible, enabling large-scale healthcare platform rollout across multiple regions
- Designed and implemented end-to-end automation workflows for infrastructure provisioning, validation, and lifecycle management, reducing manual effort by 80%+
- Developed reusable automation modules and workflows to standardize infrastructure provisioning and service delivery
- Developed scalable Python/PowerShell automation frameworks for governance, tagging, and post-provisioning validation across **1,000+** resources
- Architected secure, large-scale data transfer systems (**150M+ files**) integrating malware scanning, blob index validation, and logging for compliance
- Automated scan result validation and transfer gating, ensuring secure and compliant data movement across environments
- Resolved complex Terraform state and drift issues, improving deployment consistency and reliability across environments
- Optimized Ansible orchestration workflows for multi-stage deployments, ensuring consistent and failure-free provisioning across **500+ servers**
- Led incident response and root-cause analysis for infrastructure and performance issues across AWS and Azure environments
- Enhanced observability and monitoring using CloudWatch and Log Analytics, improving system reliability and MTTR
- Collaborated with DevOps and security teams on IAM, access control, and governance automation

Cloud Infrastructure Engineer GenPx Inc Irvine, Texas
Dec 2022 – Sept 2024

- Designed and maintained a **secure landing zone architecture for large-scale file migrations**, integrating Azure Storage, Defender for Storage malware scanning, and Log Analytics, with automated validation and controlled data movement into internal environments
- Designed and modularized Terraform and CloudFormation templates to provision VPCs, IAM roles, compute, and storage across AWS and Azure environments
- Built GitHub Actions and Jenkins pipelines to automate infrastructure deployments, including plan/apply workflows, validation checks, and rollback strategies
- Implemented centralized monitoring and alerting pipelines using CloudWatch, Azure Monitor, and Log Analytics with custom dashboards and alerts
- Developed Python-based automation for infrastructure health checks, reporting, and compliance validation
- Supported AKS/EKS cluster operations, resolving pod failures, networking issues, and logging gaps using integrated observability tools
- Improved reliability by designing alerting thresholds, dashboards, and runbooks, reducing incident resolution time

Software Engineer Cognizant Technology Solutions Hyderabad, India
Dec 2021 – Aug 2022

- Developed **AWS Lambda-based automation workflows** integrated with S3 and CloudWatch for event-driven processing and monitoring
- Built backend microservices using Go and Spring Boot, optimizing API response times and improving system performance by 25%
- Implemented CI/CD pipelines using Jenkins and Docker, enabling automated build, test, and deployment workflows
- Developed Python-based log analysis tools to parse CloudWatch logs and detect anomalies, reducing incident investigation time
- Managed and troubleshot AKS clusters, resolving pod lifecycle issues, container crashes, and networking misconfigurations
- Automated AWS infrastructure provisioning using CloudFormation templates, ensuring consistent and repeatable deployments

PROJECTS

Automated File Transfer & Validation Framework (Azure)

- Engineered an automated file transfer pipeline using **AzCopy** with validation workflows leveraging Blob Index Tags and logging
- Implemented **pre-transfer validation and monitoring**, ensuring data integrity and traceability across storage containers
- Simulated large-scale enterprise data movement workflows with secure and controlled transfer logic

Access Control Automation Platform (AWS)

- Developed a **Terraform-based IAM automation system** with OIDC integration for secure, scalable multi-account access management
- Automated role provisioning and policy enforcement, improving governance and reducing manual access management overhead

EDUCATION

Master of Science in Computer Science Stevens Institute of Technology — Hoboken, NJ

- GPA: 3.78/4 | Provost Master's Scholarship Recipient
- Relevant Coursework: Cloud Computing, Distributed Systems, Systems Design, Database Systems

Bachelor of Technology in Electronics and Communication Engineering Jain University — Bengaluru, India

- GPA: 7.8/10 | Minor in Computer Science
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks

AWARDS & ACHIEVEMENTS

- Top 0.1% in National Math Olympiad (India, 2015)
- Runner-up – JU Hackathon (2019)
- Served as **Vice President of the JT Tech Club**, leading the organization and successfully organizing multiple hackathons and technical events that fostered innovation and community engagement among members.
- Achieved Level 6 certification in Abacus with multiple regional/national competition participation